Depth-First Search (DFS) — Python Code

```
def dfs(graph, start, goal):
    stack = [start]
    visited = []
    parent = {start: None}

    while stack:
        node = stack.pop()

        if node not in visited:
            visited.append(node)
            print("Visiting:", node)

            if node == goal:
                path = []
                while node is not None:
                    path.append(node)
                    node = parent[node]

                path.reverse()
                return path

            for child in reversed(graph[node]):
                if child not in visited:
                    if child not in parent:
                        parent[child] = node
                    stack.append(child)

    return None


graph = {
    'A': ['B', 'C'],
    'B': ['A', 'D', 'E'],
    'C': ['A', 'F', 'G'],
    'D': ['B'],
    'E': ['B', 'H'],
    'F': ['C'],
    'G': ['C', 'I'],
    'H': ['E'],
    'I': ['G']
}

start = 'A'
goal = 'I'

path = dfs(graph, start, goal)

if path:
    print("Path:", " -> ".join(path))
    print("Path cost:", len(path) - 1)
else:
    print("No path exists")
```